Using the Onsen UI with Monaca IDE for Hybrid Mobile App Development

Today, a mobile app is a must for organisations that want to ensure a strong presence in a challenging market. There are different popular mobile operating systems and each has its own user base. To develop mobile applications separately for all these different operating systems and maintain them requires a lot of resources, in terms of both time and cost. Hence, many organisations are shifting towards cross-platform mobile application development. Onsen UI is a framework that enables such app development.

Cross-platform mobile apps work on multiple operating systems with a single code base. There are two types of cross-platform apps.

Native cross-platform apps: Most mobile operating systems have their own SDKs to create mobile apps. These work on specific programming languages. Android developers, for example, prefer the Java language while developers working on iOS prefer the Objective C and Swift programming languages. In cross-platform app development, we use the APIs provided by the native SDK in another programming language which is not supported by other vendors. With unified APIs, code written in one particular language will work on multiple platforms -- that's how the native cross-platform app works.

In this kind of approach, the final app still uses the native APIs, and it achieves nearly the same performance as native apps. Nowadays, the GUI plays a major role in mobile app development. It is somewhat complex to implement rich GUIs using unified APIs in native cross-platform apps. For this reason, many developers prefer hybrid HTML5 cross-platform apps.

Column1	Column2	Column3
Features	Native Apps	Hybrid Apps
Language	Native Only	HTML5,CSS,JavaScript
Cost	High	Moderate
Graphics	High	Moderate
Performance	Very Fast	Good
Portability	None	High
Device Access	Complete	Complete
Maintenance	High	Low
UI/UX	High	Moderate
Time to Market	Slower	Faster

Figure 1: Native apps vs hybrid apps

Hybrid HTML5 cross-platform apps: Almost 60 per cent of the code of any mobile application deals with the GUI. Android, iOS and Windows all have browser components in their SDKs. By leveraging these components, developers can use HTML5 to design mobile applications. So the final application is built using the native framework and HTML/JavaScript in a Web view. This is the reason why it's called a hybrid cross-platform app.

Figure 1 gives a comparison between native apps and hybrid apps.

One of the limitations of hybrid HTML5 based mobile applications is that their look and feel is not as good as that of native apps. Onsen UI is a UI framework that resolves this issue by providing a rich UI, which provides a look and feel that's very similar to that of a native app.

Onsen UI is a Web component driven UI framework for Cordova/PhoneGap apps. It supports all kinds of JS frameworks and works with AngularJS 1 and 2.

Popular hybrid frameworks

There are many hybrid mobile frameworks available in the market but none of them provide exactly the same look and feel as the native apps. All do the same things based on the requirements, so developers can choose from these frameworks.

Let's take a look at how a few popular hybrid frameworks compare with each other.

Ionic: This is a new entrant, and is quick and reliable. It is based on the AngularJS framework. It can be used just for hybrid mobile applications, while a few functionalities will also work with ordinary Web applications. It does not have a very large community